Cover letter draft — Physics Letters A

Dear Editor,

Please consider **"Phase-dependent recovery and lifetime of encoded entanglement under collective detector noise"** for publication in *Physics Letters A*.

The manuscript studies a four-qubit encoding in which two register qubits experience collective detector noise. Locally prepared states have identical initial reference–register negativity and detector marginals. An exact trace-norm reduction proves an optimal phase at every time within the stated secular weak-coupling generator. A specified detector measurement and conditional correction recover a two-qubit resource, linking its negativity to distinguishability of evolved sector populations.

Weak singlet–triplet mixing gives a metastable lifetime through a slow-sector projection. Applying this resource law to established accelerated-vacuum and thermal spectra yields a leading lifetime ratio at matched local rates and an observation time maximizing the resource contrast. The work connects collective open-system dynamics with an explicit recoverable-entanglement quantity.

We distinguish the calculation from earlier local-rotation studies of cluster-state robustness, known erasure correction, singlet protection and collective-detector spectra. Optimality is restricted to the specified phase family; recovery is a stated procedure. Exact common-coupling protection is separated from the conditional weak-coupling interpretation of the finite-separation lifetime. Necessary proofs follow the main propositions. Explicit conditional states and a comparison of fast and slow recovery stages clarify the operational interpretation. Supplementary derivations, reproducible code and figure data accompany the manuscript.

Thank you for considering this submission.

Sincerely,

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